

Platelet-type Bleeding Disorder 12

(BDPLT12): A Comprehensive Disease Characterization Report

Disease: Platelet-type Bleeding Disorder 12 (BDPLT12) **MONDO ID:** MONDO:0011588 · **OMIM:** #605735 · **Category:** Mendelian (autosomal dominant) **Causal gene:** *PTGS1* (cyclooxygenase-1 / COX-1 / PGHS-1)

Summary

Platelet-type Bleeding Disorder 12 (BDPLT12) is an ultra-rare, autosomal-dominant inherited platelet **function** disorder caused by germline loss-of-function variants in *PTGS1*, the gene encoding platelet **cyclooxygenase-1 (COX-1, also called prostaglandin-endoperoxide H synthase-1, PGHS-1)**. Because COX-1 is the enzymatic target that low-dose aspirin irreversibly acetylates, a constitutional deficiency of COX-1 activity reproduces the pharmacology of chronic aspirin exposure. For this reason the disorder is classically and interchangeably known as the **"aspirin-like defect" (ALD)**. It is catalogued as MONDO:0011588 (= OMIM #605735, DOID:0111058, MeSH C567786, UMLS C2751535, MedGen 414043, GARD 0010575).

The core pathophysiology is a single, well-defined biochemical block: deficient COX-1 activity impairs the conversion of **arachidonic acid → prostaglandin endoperoxides (PGG2/PGH2) → thromboxane A2 (TXA2)** in the platelet. Because TXA2 is the autocrine/paracrine amplifier that drives the "second wave" of platelet secretion and aggregation, its loss produces a mild, lifelong mucocutaneous bleeding tendency. The laboratory hallmark is **selectively absent or markedly reduced arachidonic-acid-induced platelet aggregation** on light transmission aggregometry (LTA), with an absent secondary wave to ADP/adrenaline but preserved response to high-dose collagen and normal ristocetin-induced agglutination. Diagnosis is confirmed by *PTGS1* sequencing.

The clinical picture is generally benign. Patients present with easy bruising, epistaxis, menorrhagia, and — most characteristically — excessive or **late post-operative/post-dental bleeding** that can be the first clue to the diagnosis. Prognosis is excellent with normal life

expectancy; the principal morbidity is perioperative, postpartum, and dental hemorrhage, plus iron-deficiency anemia from menorrhagia. Management is on-demand and peri-procedural, relying on **tranexamic acid**, **desmopressin (DDAVP)**, and **platelet transfusion** for severe bleeding, together with strict avoidance of aspirin and other NSAIDs. Nine findings were confirmed across this investigation, drawing on 39 reviewed papers, ontology cross-references (Monarch/MONDO, HPO, GO, Reactome, UniProt), and model-organism data (global and platelet-specific *Ptgs1* knockout mice).

Key Findings

F001 — BDPLT12 is a platelet-type bleeding disorder caused by COX-1 (PTGS1) deficiency

BDPLT12 is definitively established as an inherited blood coagulation disorder caused by deficiency of platelet cyclooxygenase-1. Ontology cross-references from Monarch/MONDO anchor the disease identity: **MONDO:0011588 = OMIM:605735, DOID:0111058, MeSH:C567786, UMLS:C2751535, MedGen:414043, GARD:0010575**. The MONDO definition reads: *"An inherited blood coagulation disease characterized by autosomal dominant inheritance of mildly increased bleeding, platelet aggregation defect, and impaired conversion of arachidonic acid to thromboxane A2 in platelets due to deficiency in PTGS1 activity."*

Synonyms / alternative names: BDPLT12; aspirin-like defect (ALD); platelet COX-1 deficiency; platelet cyclooxygenase-1 deficiency; PGHS-1 deficiency; prostaglandin G/H synthase-1 deficiency.

Causal gene: *PTGS1* (prostaglandin-endoperoxide synthase 1 = COX-1 / PGHS-1), **HGNC:9604**, cytogenetic location **9q33.2**, gene OMIM 176805, UniProt **P23219**, Ensembl **ENSG00000095303**.

The information is derived from **aggregated disease-level resources** (OMIM, MONDO, HPO) combined with **individual patient / small-family case reports** — there is no EHR-scale patient cohort for this ultra-rare disorder.

F002 — Clinical and laboratory phenotype of the aspirin-like defect

The largest phenotypic characterization comes from Rolf et al. (2009), a study of **17 unrelated families (52 individuals)**. Impaired platelet aggregation in response to arachidonic acid (**platelet aggregation response to AA, PAR-to-AA $\leq 10\%$**) was used as the mandatory

diagnostic criterion. At least one bleeding symptom was reported by **25 of 34 (74%)** ALD patients, and a prolonged PFA-100 closure time was detected in **24 of 34 (71%)**; both correlated significantly with impaired PAR-to-AA (**P = 0.001** and **P = 0.002**, respectively). The estimated prevalence was **~0.6% among pediatric patients with suspected coagulation disorders**, and the authors emphasize that the disorder is probably underdiagnosed because of its mild bleeding phenotype.

"Aspirin-like defect (ALD) is a rare, mostly autosomal dominant inherited dysfunction of the intraplatelet arachidonic acid (AA) pathway leading to impaired thromboxane A2 signalling." — [PMID: 19036102](#)

"At least one bleeding symptom was reported by 25 (74%) ALD patients and prolonged CT was detected in 24 (71%) of the cases, both significantly correlated with impaired PAR to AA (P = 0.001 and P = 0.002, respectively)." — [PMID: 19036102](#)

A clinically instructive case is described by Salinas et al. (2014), reporting **late post-operative hemorrhage after third-molar extraction in a patient with undiagnosed COX-1 deficiency** — the first such case documented in the English literature — underscoring how the diagnosis is frequently made only after a surgical bleeding event ([PMID: 24480756](#)).

F004 — Mechanistic causal chain: PTGS1 loss of function blocks arachidonate → TXA2

PTGS1 encodes COX-1/PGHS-1, an **endoplasmic-reticulum-membrane, heme-dependent, bifunctional cyclooxygenase-peroxidase**. Curated Gene Ontology annotations define its molecular activity and localization:

GO aspect	Term	ID
Molecular function	prostaglandin-endoperoxide synthase activity	GO:0004666
Molecular function	peroxidase activity	GO:0004601
Molecular function	heme binding	GO:0020037
Biological process	cyclooxygenase pathway	GO:0019371
Biological process	prostaglandin biosynthetic process	GO:0001516
Biological process	prostanoid biosynthetic process	GO:0046457
Cellular component	endoplasmic reticulum membrane	GO:0005789
Cellular component	endoplasmic reticulum lumen	GO:0005788

The pathway (Reactome **R-HSA-2162123**, "Synthesis of Prostaglandins and Thromboxanes") proceeds:

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Membrane phospholipids
| (cytosolic phospholipase A2, cPLA2)
▼
Arachidonic acid (CHEBI:15843)
| (COX-1 cyclooxygenase activity) ← BLOCKED in BDPLT12
▼
Prostaglandin G2, PGG2 (CHEBI:27647)
| (COX-1 peroxidase activity)
▼
Prostaglandin H2, PGH2 (CHEBI:15554)
| (thromboxane synthase, TBXAS1)
▼
Thromboxane A2, TXA2 (CHEBI:15627)
| (autocrine/paracrine activation of TP receptor, TBXA2R)
▼
Gq / G12-13 signaling → granule secretion + integrin αIIbβ3 activation
▼
Secondary-wave platelet aggregation ← ABOLISHED

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Loss of COX-1 function removes TXA2 production, abolishing this **positive-feedback amplification loop**. The primary adhesion/activation machinery (GPIb-IX-V, αIIbβ3, collagen receptors) is intact, which is why high-dose collagen and ristocetin responses are preserved while AA-induced and secondary-wave aggregation fail.

F005 — PTGS1 variant spectrum: rare heterozygous causal variants including a dominant-negative N-glycosylation defect

ClinVar catalogues roughly **137 PTGS1 variants**. For the BDPLT12/aspirin-like-defect trait, most are classified **Benign, Likely benign, or VUS**, with no established recurrent Pathogenic entry — a direct reflection of the disorder's extreme rarity and reliance on single-family reports. Reported variant types are predominantly **missense**, with **nonsense** variants also catalogued (e.g., p.Arg108Ter). Origin is **germline**; the principal functional consequence is **loss of enzymatic function**.

The landmark molecular case is **Palma-Barqueros et al. (2021)**, who identified "A novel genetic variant in PTGS1 [that] affects N-glycosylation of cyclooxygenase-1 causing a dominant-negative effect on platelet function and bleeding diathesis" ([PMID: 33326144](#)). This establishes a **dominant-negative mechanism** — a mutant subunit interfering with the wild-type product — as one route to the autosomal-dominant phenotype, complementing simple haploinsufficiency. The reference transcript is **NM_000962.4**.

"A novel genetic variant in PTGS1 affects N-glycosylation of cyclooxygenase-1 causing a dominant-negative effect on platelet function and bleeding diathesis." — [PMID: 33326144](#) (title quote)

F006 — Mouse models recapitulate the arachidonate-induced aggregation defect

Model-organism evidence strongly supports the causal mechanism. Global **Ptgs1^{-/-} mice** (Langenbach et al., 1995, *Cell* 83:483-492) are viable and show markedly reduced platelet aggregation to arachidonic acid, decreased indomethacin-sensitive prostaglandin synthesis, and reduced inflammatory/pain responses. **Platelet/megakaryocyte-specific deletions** (Pf4-ΔCre and Gp1ba-ΔCre × Cox-1^{flox/flox}) confirm a **cell-autonomous** role of platelet COX-1 (Tang et al., 2024):

"Ex vivo platelet aggregation induced by arachidonic acid or adenosine diphosphate in platelet-rich plasma was inhibited to a similar extent in [platelet-specific Cox-1-deleted mice]." — [PMID: 38660804](#)

These models phenocopy the human COX-1-deficiency aggregation defect and the pharmacology of low-dose aspirin. The mouse ortholog is *Ptgs1* (NCBI Taxon 10090).

F007 — HPO phenotype spectrum of BDPLT12

Official HPO annotations for **OMIM:605735** (JAX) define the phenotype spectrum:

Phenotype	HPO term	Type
Impaired platelet aggregation	HP:0003540	Laboratory abnormality
Bruising susceptibility / easy bruising	HP:0000978	Clinical sign
Epistaxis	HP:0000421	Symptom
Menorrhagia	HP:0000132	Symptom
Gastrointestinal/intestinal bleeding	HP:0002584	Clinical sign
Joint hemorrhage (hemarthrosis)	HP:0005261	Clinical sign
Congenital onset	HP:0003577	Onset modifier
Autosomal dominant inheritance	HP:0000006	Inheritance

Severity is characteristically **mild**; bleeding is **episodic/provoked** (surgery, dental extraction, menses, trauma) rather than spontaneous or progressive. Frequencies are not quantified in HPO, but bleeding symptoms were reported in ~74% of aspirin-like-defect patients (Rolf 2009). Phenotype type spans a **laboratory abnormality** (impaired AA-induced aggregation) plus **clinical signs/symptoms** (mucocutaneous bleeding).

F008 — Diagnostic approach: bleeding assessment + LTA (absent AA response) confirmed by PTGS1 sequencing

Hoepner et al. (2025) propose a **four-step diagnostic approach** to inherited platelet function defects: (1) history plus a validated bleeding score (**ISTH-BAT**); (2) exclusion of plasmatic coagulation disorders and von Willebrand disease; (3) platelet phenotype/function testing — blood smear light microscopy, **light transmission aggregometry (LTA)**, flow cytometry, and lumiaggregometry; (4) genetic testing.

"Established methods consist of blood smear analysis by light microscopy, light transmission aggregometry, and flow cytometry." — PMID: 39870109

"We strongly advocate for the use of a validated bleeding score like the ISTH-BAT (International Society on Thrombosis and Haemostasis Bleeding Assessment Tool)." — PMID: 39870109

For the COX-1/aspirin-like defect specifically, the LTA hallmark is **absent/markedly reduced aggregation to arachidonic acid** with a characteristically **absent secondary wave to ADP/adrenaline**, while primary response to high-dose collagen and ristocetin-induced agglutination are preserved. Biochemical confirmation is **reduced serum thromboxane B2** and **urinary 11-dehydro-thromboxane B2**. Genetic confirmation is by *PTGS1* single-gene testing or a hereditary-platelet-disorder NGS panel / WES (e.g., ThromboGenomics-type panels).

F009 — Epidemiology, inheritance, and prognosis: ultra-rare AD trait, normal life expectancy, mild lifelong course

There is **no dedicated Orphanet prevalence code**; BDPLT12 is ultra-rare with only single families/cases reported in OMIM and the literature. The broader aspirin-like-defect phenotype was estimated at **~0.6% among pediatric patients referred for suspected coagulation disorders** (Rolf 2009) and is thought to be underdiagnosed:

"Due to the mild bleeding symptoms, ALD is probably underdiagnosed." — PMID: 19036102

Inheritance is **autosomal dominant** (HP:0000006) with germline *PTGS1* variants; **penetrance appears incomplete and expressivity variable** (mild ALD subgroups with PAR-to-AA of 19–32% observed within families). There is **no genetic anticipation** (not a repeat-expansion disorder). There is no strong sex predilection for the molecular defect, although menorrhagia makes bleeding more clinically apparent in women. Prognosis is excellent: **normal life expectancy**; disease is chronic/lifelong but mild, episodic and provoked; the main morbidity is perioperative/postpartum/dental hemorrhage and iron-deficiency anemia from menorrhagia; disease-attributable mortality is negligible with appropriate hemostatic management.

F003 — Management of the COX-1/aspirin-like platelet defect

Bargehr, Knöfler & Streif (2023) review inherited platelet disorder (IPD) management; the established options apply directly to COX-1 defect:

"Established treatment options of IPDs include local hemostatic treatment, tranexamic acid, desmopressin, platelet concentrates, and recombinant activated factor VII. Hematopoietic stem cell therapy is a curative approach for selected patients." — [PMID: 37611608](#)

Peri-partum and peri-operative management of functional platelet disorders relies on **DDAVP, tranexamic acid, prophylactic oxytocics, and platelet transfusion** (well documented in related storage-pool disorders). Emerging approaches include **autologous HSC gene therapy** and artificial platelets/nanoparticles; hematopoietic stem cell transplantation is curative but reserved for the most severe (generally non-COX-1) platelet disorders.

Section-by-Section Detail (Research Template)

1. Disease Information

- **Overview:** Autosomal-dominant inherited platelet function disorder = constitutional COX-1 deficiency ("aspirin-like defect") causing mild mucocutaneous bleeding.
- **Identifiers:** MONDO:0011588; OMIM #605735; DOID:0111058; MeSH C567786; UMLS C2751535; MedGen 414043; GARD 0010575. No specific ICD-10/11 code (falls under D69.1 "qualitative platelet defects").
- **Synonyms:** BDPLT12; aspirin-like defect; platelet COX-1 / cyclooxygenase-1 deficiency; PGHS-1 deficiency.
- **Source of information:** Aggregated disease-level ontologies + individual case/family reports.

2. Etiology

- **Causal factor:** Germline heterozygous loss-of-function variants in *PTGS1* (genetic; monogenic).

- **Genetic risk factors:** *PTGS1* causal variants (missense predominant; nonsense e.g. p.Arg108Ter; N-glycosylation-disrupting dominant-negative allele). No known susceptibility loci or modifier genes are established.
- **Environmental risk factors / gene–environment interaction:** Concurrent aspirin/NSAID exposure additively worsens the phenotype (pharmacologic COX-1 inhibition on top of genetic deficiency) — the clearest gene–environment interaction; other antiplatelet drugs likewise unmask/aggravate bleeding.
- **Protective factors:** None specifically documented; avoidance of antiplatelet agents mitigates risk.

3. Phenotypes

See F007 table. Onset congenital; severity mild; course stable/episodic-provoked; laboratory abnormality (impaired AA aggregation) is the most penetrant feature. Quality-of-life impact is generally low, dominated by menorrhagia-related anemia and perioperative bleeding anxiety; no disease-specific QoL instrument data exist.

4. Genetic / Molecular Information

See F001, F005. Gene *PTGS1* (HGNC:9604, 9q33.2); reference transcript NM_000962.4; protein UniProt P23219. Variant classes: missense, nonsense; germline origin; loss-of-function and dominant-negative consequences. Most ClinVar entries VUS/benign. No established modifier genes, epigenetic mechanisms, or chromosomal abnormalities for this disorder.

5. Environmental Information

Non-genetic contributors are limited to **pharmacologic COX inhibitors** (aspirin, NSAIDs) that phenocopy or aggravate the defect. No infectious agents, toxins, or lifestyle factors cause the disorder.

6. Mechanism / Pathophysiology

See F004 and the Mechanistic Model below — causal chain from *PTGS1* variant to defective TXA2-dependent second-wave aggregation.

7. Anatomical Structures Affected

- **Organ/system:** Hematopoietic and hemostatic system; primary "organ" is the circulating platelet.

- **Cell types:** Platelet (CL:0000233); megakaryocyte (CL:0000556, site of COX-1 synthesis).
- **Subcellular:** ER membrane (GO:0005789) where COX-1 resides.
- **Localization of bleeding:** Mucocutaneous surfaces — skin (UBERON:0002097), nasal mucosa (UBERON:0001825), endometrium/uterus (UBERON:0000995); occasionally GI tract and joints. Bleeding is systemic/bilateral, not lateralized.

8. Temporal Development

Congenital onset (HP:0003577); insidious/chronic; lifelong and stable (non-progressive); manifestations episodic and provoked by hemostatic challenge. No disease stages, no remission/relapse cycles; critical periods are surgical, dental, obstetric, and menstrual events.

9. Inheritance and Population

See F009. Autosomal dominant, incomplete penetrance, variable expressivity, no anticipation; ultra-rare with no reliable prevalence estimate; no founder effect or consanguinity role documented (AD, not AR).

10. Diagnostics

See F008. Core: ISTH-BAT bleeding score → exclude VWD/coagulation factor defects → LTA (absent AA response, absent second wave) → serum TXB2 / urinary 11-dehydro-TXB2 → *PTGS1* sequencing (single-gene or NGS platelet-disorder panel/WES). Platelet count and morphology are normal.

11. Outcome / Prognosis

Excellent; normal life expectancy; negligible disease-specific mortality. Morbidity = perioperative/postpartum/dental hemorrhage and iron-deficiency anemia from menorrhagia. Prognostic factor: residual COX-1 activity (PAR-to-AA level) correlates with bleeding tendency.

12. Treatment

See F003. Pharmacotherapy/procedural: tranexamic acid (NCIT antifibrinolytic), desmopressin/DDAVP, platelet concentrates, recombinant activated factor VIIa for refractory bleeding; local hemostatic measures. Pharmacogenomic caution: strict avoidance of aspirin/NSAIDs. Advanced/experimental: autologous HSC gene therapy and engineered/artificial platelets are emerging but not standard for this mild disorder.

13. Prevention

No primary prevention (genetic). Secondary/tertiary prevention = pre-procedure hemostatic planning, avoidance of antiplatelet drugs, treatment of iron deficiency, and **genetic counseling** for AD inheritance with variable penetrance. Cascade testing of at-risk relatives is appropriate once a family variant is identified.

14. Other Species / Natural Disease

Mouse ortholog *Ptgs1* (NCBI Taxon 10090). No naturally occurring companion-animal BDPLT12 equivalent is catalogued in OMIA; the disease is understood chiefly through engineered mouse models (see F006). COX-1/prostanoid biology is evolutionarily conserved across mammals.

15. Model Organisms

Global *Ptgs1*^{-/-} mice (Langenbach 1995) and platelet/megakaryocyte-specific conditional knockouts (Pf4-ΔCre, Gp1ba-ΔCre × Cox-1^{flox/flox};

P 38660804

P 31248980

recapitulate the AA-induced aggregation defect and reduced platelet prostanoid biosynthesis — strong phenotype recapitulation of the human loss-of-function state. **Limitation:** these model complete loss/haploinsufficiency, not the specific human dominant-negative N-glycosylation allele; a knock-in model is lacking.

Mechanistic Model / Interpretation

BDPLT12 is a textbook example of a **single-enzyme, single-pathway platelet function disorder** in which the clinical phenotype maps cleanly onto a defined biochemical lesion. The causal chain, from mutation to bleeding, is:

1. Germline heterozygous *PTGS1* variant (missense, nonsense, or N-glycosylation-disrupting)
| leads to
2. Reduced or dysfunctional COX-1 protein in megakaryocytes/platelets
| (haploinsufficiency OR dominant-negative interference with wild-type subunit)
| results in
3. Impaired cyclooxygenase conversion: arachidonic acid $\nrightarrow$ PGG2/PGH2
| results in
4. Deficient thromboxane A2 (TXA2) synthesis ($\downarrow$ serum TXB2, $\downarrow$ urinary 11-dehydro-TXB2)
| results in
5. Loss of TXA2/TP-receptor autocrine amplification of platelet activation
| results in
6. Absent secondary wave of secretion & aggregation; selectively absent AA-induced aggregation
| results in
7. Impaired primary hemostasis (defective platelet plug formation at sites of injury)
| manifests as
8. Mild, provoked mucocutaneous bleeding: bruising, epistaxis, menorrhagia,
and late post-surgical/post-dental hemorrhage

Upstream vs downstream. The initiating lesion (steps 1–2) is the *PTGS1* variant and reduced COX-1 protein. The proximate biochemical defect (steps 3–4) is the arachidonate→TXA2 block. The downstream physiological consequence (steps 5–7) is loss of the amplification loop and defective platelet plug formation, and the clinical manifestation (step 8) is the mild bleeding tendency.

Two molecular routes to autosomal dominance. A heterozygous variant can cause disease either by (a) **haploinsufficiency** — 50% enzyme is insufficient for full second-wave amplification under stress — or (b) **dominant-negative interference**, as demonstrated for the N-glycosylation-disrupting variant (Palma-Barqueros 2021), where the mutant subunit impairs function beyond simple dose reduction. Variable penetrance/expressivity (PAR-to-AA ranging 19–32% in mild subgroups) is consistent with residual COX-1 activity determining phenotype severity.

Cell types and biological processes. The affected cell is the **platelet** (CL:0000233) and its precursor the **megakaryocyte** (CL:0000556); COX-1 protein synthesis occurs primarily in megakaryocytes and is loaded into circulating platelets. The relevant biological processes are the **cyclooxygenase pathway** (GO:0019371) and **prostanoid/thromboxane biosynthesis** (GO:0046457, GO:0001516), localized to the **ER membrane** (GO:0005789).

Distinguishing feature vs other platelet disorders. Unlike Glanzmann thrombasthenia (α Ib β 3 defect, absent aggregation to all agonists), Bernard-Soulier syndrome (GPIb-IX-V, macrothrombocytopenia), and platelet-type von Willebrand disease (GP1BA gain-of-function, platelet **hyper**responsiveness), BDPLT12 shows a **selective** aggregation defect confined to the

AA/TXA2 pathway with normal platelet count and morphology. The single most discriminating test is the isolated absence of AA-induced aggregation with preserved high-dose collagen and ristocetin responses.

Disorder	Gene	Defect type	Platelet count	Aggregation pattern
BDPLT12 (COX-1 def.)	<i>PTGS1</i>	LoF / dominant-negative	Normal	Selective loss of AA response; absent 2nd wave
Glanzmann thrombasthenia	<i>ITGA2B/ITGB3</i>	LoF	Normal	Absent to all agonists (except ristocetin)
Bernard-Soulier syndrome	<i>GP1BA/GP1BB/GP9</i>	LoF	Low (large platelets)	Absent ristocetin agglutination
Platelet-type VWD	<i>GP1BA</i>	Gain-of-function	Low	Enhanced low-dose ristocetin

Evidence Base

PMID	Title (abbrev.)	Evidence type	Supports
19036102	<i>Clinical and laboratory phenotypes associated with the aspirin-like defect (17 families)</i>	Human clinical cohort	F002, F009 — bleeding frequency (74%), PFA-100 (71%), prevalence ~0.6%, AD inheritance, underdiagnosis
33326144	<i>Novel PTGS1 variant affects N-glycosylation of COX-1, dominant-negative effect</i>	Human genetics + functional	F005 — dominant-negative molecular mechanism
24480756	<i>Late postoperative hemorrhage in undiagnosed COX-1 deficiency after third molar extraction</i>	Human case report	F002 — surgical/dental bleeding presentation
38660804	<i>Pf4-ΔCre vs Gp1ba-ΔCre depletion of COX-1 in platelets</i>	Model organism (mouse)	F006 — cell-autonomous AA-induced aggregation defect
39870109	<i>Diagnostic assessment of inherited platelet function defects, Part 1</i>	Clinical methodology review	F008 — LTA, flow cytometry, ISTH-BAT diagnostic pathway
37611608	<i>Treatment of Inherited Platelet Disorders: Current Status and Future Options</i>	Treatment review	F003 — tranexamic acid, DDAVP, platelet concentrates, HSCT
16684008	<i>Congenital platelet disorders: mechanisms, diagnosis, treatment</i>	Review	Background — second-wave aggregation defects, management
16102044	<i>Qualitative disorders of platelets and megakaryocytes</i>	Review	Background — differential diagnosis landscape
31248980	<i>Platelet-specific deletion of COX-1 ameliorates DSS colitis</i>	Model organism	F006 — platelet COX-1 conditional KO recapitulates low-dose-aspirin pharmacology
26272103	<i>Abnormal megakaryopoiesis and platelet function in COX-2-deficient mice</i>	Model organism	Contrast — COX-2 vs COX-1 roles in platelets

How the evidence fits together. The human cohort (P 19036102)

establishes the clinical/laboratory phenotype and epidemiology; the molecular case report (P 33326144)

provides the mechanistic proof that a *PTGS1* variant is causal and defines a dominant-negative route; the mouse conditional-knockout studies (PMIDs 38660804, 31248980) supply cell-autonomous causal evidence that platelet COX-1 loss reproduces the exact aggregation defect; the diagnostic and treatment reviews (PMIDs 39870109, 37611608) translate the mechanism into clinical practice. No reviewed paper challenges the core model; the main tension in the literature is between COX-1 and COX-2 contributions in *vascular* (not platelet) prostanoid balance (PMIDs 31510878, 27020548), which is peripheral to the platelet-intrinsic BDPLT12 phenotype.

Limitations and Knowledge Gaps

1. **Genotype–phenotype ambiguity.** No recurrent pathogenic *PTGS1* variant is established for BDPLT12; most ClinVar entries are VUS/benign. ACMG-grade classification of causal variants is limited by absence of segregation and functional data for most alleles.
2. **Nomenclature and cohort mixing.** The literature conflates the clinically defined "aspirin-like defect" (a functional label that can arise from several upstream signaling or thromboxane-pathway defects) with genetically confirmed *PTGS1* deficiency (BDPLT12 proper). Some ALD cohort statistics (e.g., the 74% bleeding frequency) derive from functionally defined patients not all genotyped for *PTGS1*.
3. **No epidemiological denominator.** There is no Orphanet prevalence code or registry; true prevalence, carrier frequency, and population/geographic distribution are unknown. The ~0.6% figure is restricted to a pediatric referral population and is not a general-population estimate.
4. **Incomplete penetrance/expressivity not mechanistically resolved.** Modifier genes, environmental factors (concurrent NSAID use), and residual-activity thresholds that determine bleeding risk are undefined.
5. **No disease-specific outcome data.** Quality-of-life instruments (EQ-5D, SF-36, PROMIS), long-term natural-history cohorts, and treatment-response rates specific to BDPLT12 are

unpublished; management is extrapolated from the broader inherited-platelet-disorder literature.

6. **No animal model of a human dominant-negative allele.** Mouse data derive from complete/conditional Cox-1 knockouts, which model loss-of-function/haploinsufficiency but not the N-glycosylation dominant-negative human variant.

Proposed Follow-up Experiments / Actions

1. **Curate a *PTGS1*–BDPLT12 variant registry** linking each reported variant to LTA phenotype (PAR-to-AA), serum TXB2, and bleeding score (ISTH-BAT), enabling ACMG reclassification and genotype–phenotype correlation.
 2. **Functional validation of VUS** in a heterologous COX-1 expression system (PGH2/TXA2 output assay) and in patient- or iPSC-derived megakaryocytes, to distinguish loss-of-function, dominant-negative, and benign alleles.
 3. **Prospective phenotyping study** genotyping all functionally defined ALD patients for *PTGS1*, to separate true BDPLT12 from phenocopies and derive genotype-specific bleeding frequencies and prevalence.
 4. **Standardize a confirmatory biochemical panel:** pair LTA (AA agonist) with serum TXB2 and urinary 11-dehydro-TXB2 cutoffs to build a reproducible diagnostic algorithm distinguishing COX-1 deficiency from thromboxane-synthase and TP-receptor defects.
 5. **Knock-in mouse of a human dominant-negative *Ptgs1* allele** (e.g., an N-glycosylation-disrupting variant) to test whether it reproduces the AD phenotype beyond haploinsufficiency and to serve as a therapeutic testing platform.
 6. **Treatment-outcome case series** in genetically confirmed BDPLT12, documenting response to tranexamic acid, DDAVP, and platelet transfusion, including perioperative and obstetric outcomes.
 7. **Genetic counseling framework** for AD inheritance with incomplete penetrance: cascade testing and pre-procedure risk assessment for affected families.
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Ontology Term Index

- **Disease:** MONDO:0011588; OMIM #605735; DOID:0111058; MeSH C567786; UMLS C2751535

- **Gene/Protein:** *PTGS1* HGNC:9604; UniProt P23219; Ensembl ENSG00000095303; gene OMIM 176805; mouse *Ptgs1*, NCBI Taxon 10090
- **GO (BP):** GO:0019371 (cyclooxygenase pathway); GO:0001516 (prostaglandin biosynthesis); GO:0046457 (prostanoid biosynthesis)
- **GO (MF):** GO:0004666 (prostaglandin-endoperoxide synthase activity); GO:0004601 (peroxidase); GO:0020037 (heme binding)
- **GO (CC):** GO:0005789 (ER membrane); GO:0005788 (ER lumen)
- **CL:** CL:0000233 (platelet); CL:0000556 (megakaryocyte)
- **UBERON:** UBERON:0002097 (skin); UBERON:0001825 (nasal mucosa); UBERON:0000995 (uterus)
- **CHEBI:** 15843 (arachidonic acid); 27647 (PGG2); 15554 (PGH2); 15627 (thromboxane A2)
- **HPO:** HP:0003540, HP:0000978, HP:0000421, HP:0000132, HP:0002584, HP:0005261, HP:0003577, HP:0000006
- **Reactome:** R-HSA-2162123 (Synthesis of Prostaglandins and Thromboxanes)
- **NCIT (treatment):** tranexamic acid, desmopressin, platelet transfusion, recombinant factor VIIa

Report compiled from 9 confirmed findings and 39 reviewed papers across a 5-iteration autonomous investigation. Evidence types are labeled human clinical, model organism, in vitro, or computational/ontology throughout.
